Hung V. Vu

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Research Interests

Nonequilibrium statistical mechanics, rare-event estimation, stochastic thermodynamics

Education

University of Science and Technology of Hanoi — Hanoi, Vietnam
Major: Semiconductor and Integrated Circuit Technology — 10/2024 – 07/2027 (expected)
GPA: 3.5
Relevant Coursework: Statistical Mechanics, Mathematics for Physics, Electronics, Semiconductor Physics

National Tsing Hua University (Exchange) — Hsinchu, Taiwan
09/2026 – 01/2027

Research Experience

Institute of Atomic and Molecular Science, Academia Sinica — Taipei, Taiwan
Research Assistant, Quantum Theory of Advanced Materials — 10/2026 - Present
Worked with Prof. Mei-Yin Chou and Dr. Martin Callsen

Institute of Physics, Vietnam Academy of Science and Technology (IOP-VAST) — Hanoi, Vietnam
Research Assistant, Biophysics Group — 12/2025 – Present
Worked with Assoc. Prof. Trinh Xuan Hoang in protein folding and stochastic thermodynamics

- **Rare-Event Sampling & Normalizing Flows:** Implementing tail-aware normalizing flows in JAX to augment importance sampling for Jarzynski free energy estimation. Focus on variance reduction in rare-trajectory regimes via learned proposal distributions.
- **Computational Physics (C++/JAX):** Developed high-performance simulations of nonequilibrium work measurements using BAOAB integrator; parallelized ensemble sampling with OpenMP on IOP's HPC cluster. Convergence analysis via phase diagrams over protocol speed and system size.
- **Go-Model Protein Folding:** Implemented parallel exact enumeration of 2D lattice protein structures and free energy landscapes. Optimized C++ enumeration algorithms for rapid structure classification.
- **Numerical Methods:** Expertise in stochastic integration, importance sampling, MATLAB/Mathematica symbolic computation, and convergence diagnostics for rare-event estimators.

Physics and Young Magazine -Vietnam Physics Society — Hanoi, Vietnam
Research Assistant, Optics Group — 10/2025 – 8/2026

- **Numerical Ray Tracing (MATLAB/C++):** Engineered high-precision ray-tracing simulations to solve Euler-Lagrange differential equations for 3D laser propagation in graded-index media. Extracted Fermat ray invariants across spherical refractive-index profiles.
- **Variational Methods:** Formulated Fermat's principle-based frameworks to model complex optical dynamics and predict beam collimation in Luneburg-Hooke profiles.

Publications

In Revision

Tran Ky Vi, Nguyen Viet Hung, Pham Van Hai, Le Viet Hoang, **Vu Viet Hung**, Nguyen Pham Long Nhat, Phan Huy Thanh Dat, Bui Ha Trang, Le Xuan The Tai, and Thanh D. Pham. "Unified Optical–Mechanical Ray Dynamics in Spherical GRIN Media." *Journal of Optics*, under revision.

In Preparation

1. Hung V. Vu and Trinh Xuan Hoang. "Convergence Analysis and Phase Diagram of the Jarzynski Equality in Langevin Dynamics with BAOAB Integration."

Technical Expertise

Computational & Programming: Python, JAX (neural networks, autodiff, JIT compilation), C++ (performance-critical simulations, OpenMP parallelization), MATLAB (numerical analysis, visualization), Mathematica (symbolic computation)

Scientific Computing: Stochastic integration (Langevin dynamics, BAOAB), rare-event sampling, importance sampling, normalizing flows, variance reduction techniques, convergence analysis, HPC (OpenMP, cluster computing)

Simulation & Modeling: Molecular dynamics, protein folding (lattice models), free energy estimation, ray tracing, nonlinear differential equations (Euler-Lagrange)

Software & Tools: COMSOL, Quantum ESPRESSO, Wannier90, PERTURBO, Cadence, Proteus, LTSpice

Languages: Vietnamese (native), English (fluent), French (intermediate), Chinese (beginner)

References

Prof. Trinh Xuan Hoang
Institute of Physics, Vietnam Academy of Science and Technology (IOP-VAST)